

Synpo Performance Diagnostic Protocol

Purpose: identify why manual corrections can take 10-15 minutes on one computer while the same data and software run normally elsewhere.

Protocol for Synpo 0.8.x. Use a duplicate project or a test specimen so the prescribed corrections do not alter the production review record.

1. What Synpo records

Open Advanced -> Diagnostic level -> Full-session diagnostics, then enable Advanced -> Diagnostic mode. Synpo creates a timestamped folder below:

<project output>\Synpo diagnostics\<timestamp>\

- events.jsonl - operation and correction phase timings.
- system-samples.csv - two-second CPU, memory, and disk-I/O samples.
- environment.json - operating system, hardware, package versions, and process names.
- session-summary.json - elapsed time and aggregated phase durations.
- README.txt - privacy and sharing notes.

No microscopy pixels, project comments, command lines, or window titles are collected. Home-directory prefixes are replaced with %USERPROFILE%. Review the JSON and CSV files before sharing them.

2. Prepare a controlled comparison

1. Use the same Synpo version, Conda environment, source TIFF pair, project settings, and correction sequence on the work PC and one normally performing PC.
2. Use a duplicate project/output directory for the experiment.
3. Connect laptop power. Record Windows power mode and whether an external monitor or Remote Desktop is in use.
4. Wait for automatic detection to finish. Close unrelated CPU-, memory-, or disk-heavy applications.
5. Confirm correction memory strategy is Automatic.
6. Start Full-session diagnostics before opening the test project.

3. Run the correction sequence

Use one detected specimen. Note the wall-clock duration and the exact status message after every step.

Step | Action | What it isolates | Duration / notes

- 1 | Individual Z-slice mode: accept a small object. | First editable-mask initialization plus metadata checkpoint. |
- 2 | Accept a second small object. | Recurring metadata/checkpoint cost without initialization. |
- 3 | Add one small missed object. | Bounded local segmentation and undo write. |
- 4 | Exclude one small spine. | Full-volume object lookup plus a small object patch. |
- 5 | Exclude or split one large dendrite. | Full-volume lookup, large patch computation, compression, and write. |
- 6 | Switch to drawable XY maximum projection and repeat a small status or mask action. | Full projection selection and post-correction projection regeneration. |
- 7 | If scientifically safe in the duplicate, perform one projection transfer. | Full-Z reads, two undo volumes, and full-Z writes. |

For every correction, record whether Synpo reports Slow low-memory correction was used. Also record whether Preparing editable masks appears after the first correction; it normally should not.

4. Observe Windows while a correction is slow

Task Manager

Open Task Manager -> Details or Performance. During the correction, record:

- Python/Synpo CPU percentage and whether one logical CPU is saturated.

- Current CPU speed compared with the base speed.
- Total memory use, Synpo working set, and committed memory.
- Disk active time, transfer rate, and response time.
- Activity from MsMpEng.exe, OneDrive, backup, DLP, indexing, or other security software.

Resource Monitor

Open Resource Monitor -> Disk. Select the Python/Synpo process and any active security or synchronization processes. Record:

- Which process accesses .synpo-cache, review.zarr, undo, or .synpo-review-*.
- Disk response time and queue length.
- Whether active time is near 100% despite low throughput.

Optional: Process Monitor with IT approval

Capture only the slow operation. Filter to the Python process and paths containing .synpo-cache. Do not collect unrelated activity longer than necessary. Look for repeated access by antivirus/security filters, long file-operation durations, sharing violations, and repeated retries.

5. Test the cache location

The source TIFF being local is not sufficient: corrections primarily use the project output cache. Create one duplicate test whose output is a plain local SSD folder such as C:\SynpoBench. Keep the original data and settings unchanged.

Check whether the original output path is under Documents, Desktop, OneDrive, Dropbox, SharePoint, a mapped drive, or a corporate redirected folder. On Windows, IT can inspect a suspected path with:

fsutil reparsepoint query "C:\path\to\output"

Record drive type, free space, BitLocker status if known, and whether the folder is synchronized or backed up. Do not change security settings merely to complete this comparison.

6. Repeatability matrix

Computer | Output/cache location | Detection running? | View mode | Operation | Processing mode | Seconds

Work PC | | Z slice | First accept | |

Work PC | | Z slice | Second accept | |

Work PC | | Z slice | Small add | |

Work PC | | Z slice | Small exclude | |

Work PC | | XY maximum | Small correction | |

Reference PC | | Z slice | Same sequence | |

7. Interpret the evidence

Finding | Likely cause

editable_masks_initialization_finished dominates only the first correction. | Normal one-time editable-mask creation; optimize initialization only if excessive.

Editable-mask initialization occurs repeatedly for the same specimen. | Review cache is incomplete, rejected, deleted, or mismatched to the detection signature.

full_volume_object_bbox_search_finished dominates. | Repeated whole-volume scanning; object extent metadata/indexing should be optimized.

undo_checkpoint_write_finished or corrected_mask_write_finished dominates. | Compression, storage latency, synchronization, or endpoint scanning.

context_generation_completed dominates after each edit. | Projection regeneration; compare Z-slice mode and avoid unnecessary invalidation.

Low CPU, 100% disk active time, low throughput, high response time. | Small-file latency, security scanning, synchronization, or paging.

High CPU on one logical processor with normal disk response. | CPU-bound single-threaded correction or compression.

.synpo-review-* receives heavy traffic. | Disk-backed low-memory correction.

The same action alternates between fast and slow with different competing processes. | External contention, scheduled scanning, power/thermal state, or storage filtering.

A plain C:\SynpoBench output is fast while the normal output is slow. | Folder redirection, synchronization, security policy, or storage-path characteristics.

8. Finish and share safely

1. Select Advanced -> Finish and package diagnostic session.
2. Open the ZIP and review environment.json, events.jsonl, and session-summary.json.
3. Attach the work-PC and reference-PC ZIPs together, labeled by machine.
4. Include the completed repeatability matrix and any Task Manager/Resource Monitor observations.
5. If endpoint software appears involved, provide the bundle to IT. Request investigation or an approved narrow test-not blanket disabling of protection.

End of protocol.